

CURRICULUM VITAE

 GENERAL INFORMATION



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 SUMMARY

Dynamic Application Manager and highly accomplished leader with 10 years of experience in medical imaging, digital healthcare solution and data-driven innovation, in-depth healthcare insight and rich academic achievements in the industry as linked. Driving innovation and operational excellence in medical imaging solutions. Proven expertise in portfolio management, product development, clinical application support, and strategic marketing across global, APAC, and China markets. Strong research background in MR physics, drug delivery mechanism, AI-driven medical technologies and data science, complemented by a PhD and a track record of scientific publications and project leadership.

A proven track record of fostering collaboration among cross-functional teams to deliver high-impact clinical applications and solutions. Committed to advancing healthcare through effective communication and a focus on academic excellence. and client-centric services.

 EDUCATION BACKGROUND

Degree	Major & Direction	Institution	Date Attended	Year Awarded
Ph.D.	MR imaging & Molecular imaging application &	Chinese Academy of Science, Wuhan Insititute of	2011-2015	2015

Degree	Major & Direction	Institution	Date Attended	Year Awarded
	Development	Physics and Mathematics		
B.S.	Chemical engineering	Wuhan Institute of Technology	2003-2007	2007

🌐 **Work Experiences**

(In chronological order beginning with current position, contains dates held and full title.)

Senior Application Manager

Bayer Healthcare China (05/2024 - Present) | Guangzhou, China

- Spearheaded clinical development initiatives for Bayer radiology products, building comprehensive disease-based solutions to drive business growth, building products based disease solution as potofolio to drive business growth.
- Developed innovative business models, offering comprehensive clinical application and scientific services to clients.
- Resourcing to build clinical [digital tools](#) such as ROI Calc, Thermostat, etc for customers to boost business & AI Agentic training chatbox for improve internal communication by leveraging RAG methodology

Digital Solution Brand Management Manager

Bayer AG - Bayer Radiology Global (03/2023 - 05/2024) | Berlin, Germany

- Curated and strategic developed Bayer's Radiology digital marketing research platform, leveraging Google Cloud for enhanced business intelligence and data-driven decision-making.
- Lead the digital health portfolio strategy for AI-driven medical imaging solutions, integrating cutting-edge technology into preclinical research pipelines.
- Direct cross-functional initiatives, ensuring alignment of digital solutions with global portfolio management objectives in R&D.

Radiology Solution Marketing Associate Manager

Bayer AG - Bayer Radiology Global (08/2022 - 02/2023) | Berlin, Germany

- agile and scrum operations, streamlining processes for the global imaging workflow across multiple radiology portfolios, including Liver & Prostate chapters.
- Lead and Coordinate IAWS Pilot projects with regional commercial team across 21 countries to orchestrate the integration of software, medical devices and drug, resulting in improved operational efficiency and portfolio synergy.
- Lead and launch Liver HCC A-map clinical project, aligning product roadmaps with market needs and business objectives across APAC & China.

Radiology Marketing associate

Bayer APAC (12/2021 - 07/2022) | Singapore, Singapore

- Directed the portfolio solution design and execution for MRI brand market transformation in the APAC region, increasing market share and driving product adoption.
- Developed and implemented marketing strategies that aligned with regional market demands and corporate goals, ensuring successful product launches and market penetration.

Digital Solution Manager (Concurrently)

Bayer AG - Bayer Radiology China (01/2021 - 11/2021) | Guangzhou, China

- Led the creation and execution of digital platform product roadmaps, paving the way for future product implementations in the Chinese market.
- Initiated innovative marketing strategies that positioned the company as a leader in digital solutions, resulting in increased market presence and competitive advantage.

Application Expert

Bayer AG - Bayer Radiology China (01/2018 - 11/2021) | Guangzhou, China

- scientific & technical support with regards to internal & external, initiating innovative projects for new business solution and driving sales figure increasing
- Acted as a project leader, coordinating with multiple stakeholders to deliver successful outcomes on preclinical research initiatives.
- Led customer-focused projects, enhancing service delivery and building strong client relationships.

Scientific Specialist

Siemens Healthineers (09/2015 - 12/2017) | Guangzhou, China

- Collaborate with cross-functional teams in executing translational research initiatives from post clinical trials to marketing promotion.
- Managed scientific marketing initiatives, providing technical support and driving the development of novel products.
- Led the implementation of WIP projects in China, contributing to the advancement of MRI technologies and scientific research.

Assistant Lecturer

- Dates: 08/2010-09/2011
- Institution name: *Hubei Polytechnic University, Hubei, China*
- Base: Huangshi, Hubei

 Certificate & Reputation (medical and scientific societies)

Member/officer	Name of Organization	Dates
Member	International Society for Magnetic Resonance in Medicine	2017-2020
fellow	Chinese Society of Imaging Technology	2017-2020

 Professional Strengthens & Skills:

Expertise in medical physics, biomedical engineering, including **MR Physics & Molecular imaging**, skilled at *MR sequence application & development, medical imaging post process & data analysis*, software development skills including **Data Science & AI:** AI Algorithms, Machine Learning, Deep Learning, CNN/FACN, MONAI, LLM, RAG, LangChain, **full stack programme on R, Python, Matlab, SQL** and containerized technologies by leveraging cloud computing & data lake into ,etc; **Well-trained scientific writer & researcher** with rich *scientific research experiences, publications, patents & code projects as listed below

Marketing research & figure analysis by **data sciences insights**; marketing strategies & product roadmap designing and development, based on **business-driven thinking & customer oriented perspective**

Strong collaboration capability & communication skill; clinical & technologies oriented; acclaimed customer reputation, honored as specific consultation & society fellow



PROJECT EXPERIENCES AND ACHIEVEMENTS

(Summarize [in chronological order] past research support and list the following for current extramural and intramural research achievements).

1. **Involving Global Management Graduation Program** - Selected as a high-potential talent for Bayer's 3-year global rotation program. Gained multi-functional experience across sales & marketing, R&D, service support, customer excellence, and digital solution business in different regions (2022 - 2024). Drove coordination and synchronous operation of multiple value chains, enhancing global strategic alignment.
2. **Digital Solution business**, leading & steering Chinese digital platform launch team, identify Chinese digital platform roadmap, readiness for the future product implementation; Pave the way for the global digital platforms launching and strategies. invest potential local AI products in China and build collaboration with them based on Bayer's future digital platform.
3. **AI-Powered Marketing Research** Launched marketing research and orchestrated marketplace intelligence by utilizing LLM and Generative AI models, leveraging prompt engineering for content generation and RAG/LangChain for knowledge base maximization.
4. **Radimetrics Application & Development** Responsible for the radiology's digital software, especially the radiation dose management software Radimetrics application & further development in China. Providing consultant & service support for Radimetrics to the customers; orchestrated with global digital solution team's strategies for Chinese software business launching & implementation.
5. **PAI Workflow Initiative & Dataset Building (Primovist AI Project)**: Initiated a collaborative research trial (Phase I in China) with a leading hospital and a 3rd party partner focusing on Primovist-related AI applications. This initiative resulted in 1 pitch publication at ISMRM 2020 and the generation of 4 Bayer IP Assets (10/2018 - 12/2020), demonstrating strong capabilities in evidence generation and innovation.
6. **AI-Driven Clinical Trial Co-creation**: Built AI efficiency solution for Bayer products by leveraging cutting-edge AI technologies (CNN/FACN). Proposed and initiated innovative clinical trials, with publications and presentations at international conferences.

7. **Global Digital Solution Ecosystem Curation:** Collaborated with 3rd party vendors to curate disease-oriented AI marketplaces, integrating GPT and Meta LLM2/LLM3 into routine marketing business.
8. **DNN Algorithms for Contrast Media Pharmacokinetic Research:** Led an innovative AI project in collaboration with Nanfang Hospital and Guangzhou University of Chinese Medicine to develop Deep Neural Network (DNN) algorithms for pharmacokinetic research of contrast media. Presented findings via 2 posters at Bayer R&D Science Conferences (Amsterdam, 2022), showcasing advanced data analysis and research capabilities.
9. **Cloud-Based Imaging Service:** Initiated cloud/On-prem based diagnosis imaging service and digital health service for clinical trials, collaborating with Tencent Cloud and Tencent Miying.Inc.
10. **Crossfunctional Platforming capabilities** Good relationship with Research institutes, Universities & Hospitals, IT partners to build business platform/ecosystem: cooperate with NVIDIA Inc, Tencent, some start-ups, etc. to build platform overarchingly eg. NVIDIA MONAI-based Deep Vein Thrombosis diagnosis framework collaborated with Guangzhou Medical University & Hospitals, etc.
11. **Scientific Differentiation Support:** Drove business figure increases for target clients by initiating scientific differentiation support projects. eg. Leading and developing DNN algorithms for contrast media's pharmacokinetic research -- a products related innovative AI project with Nanfang Hospital, University of Guangzhou Chinese Medicine collaboration, 2 posters on Bayer R&D Science Conferences, Amsterdam, 2022
12. **digital marketing & platform marketing** Contributed significantly to the [Liver360 project](#) for liver cancer disease awareness and medical education promotion in the APAC region (2021), supporting strategic marketing and medical communication objectives.
13. **Business driven & Commercial Insights:** As sales & application expert drove business figure above 2% MS increase for target clients by initiating scientific differentiation support project in Guangzhou, 2020 - 2021
14. **Preclinical & Post Clinical Research:** Initiated several preclinical & post clinical MR imaging & Contrast media research pilots with 50+ hospitals, generating peer-reviewed publications.

Publications

Peer reviewed papers

1. Chen Y, Li X, Song Y, **Zhu X**, Zhao J, Yan X, Wang J, Fan M, Chu J. The diagnostic efficacy of amide proton transfer imaging in grading gliomas and predicting tumor proliferation. [Neuroreport. 2019 Jan 16;30\(2\):139-144](#)
2. Cui Y, Cao Y, Song J, Dong N, Kong X, Wang J, Yuan Y, **Zhu X**, Yan X, Greiser A, Shi H, Han P. Association between myocardial extracellular volume and strain analysis through cardiovascular

- magnetic resonance with histological myocardial fibrosis in patients awaiting heart transplantation. [J Cardiovasc Magn Reson. 2018 Apr 23; 20\(1\): 25](#)
3. Wu B, Lu ST, Yu H, Liao RF, Li H, Lucie Zafitatsimo BV, Li YS, Zhang Y, **Zhu XL**, Liu HG, Xu HB, Huang SW, Cheng Z. Gadolinium-chelate functionalized bismuth nanotheranostic agent for in vivo MRI/CT/PAI imaging-guided photothermal cancer therapy. [Biomaterials. 2018 Mar;159:37-47](#)
 4. Huang WY, Li MM, Lin SM, Chen F, Yang K, **Zhu XL**, Wu G, Li JJ. In Vivo Imaging Markers for Prediction of Radiotherapy Response in Patients with Nasopharyngeal Carcinoma: RESOLVE DWI versus DKI. [Sci Rep. 2018 Oct 26;8\(1\):15861.](#)
 5. Weiyuan Huang, Wu Gang , Meng-meng Li , Feng Chen , Shao-min Lin , Kai Yang , Ying-man Zhao , **Xiao-lei Zhu** , Jian-Jun Li. [Oncotarget 2017 Aug 2;8\(38\)](#)
 6. Wu G, Xie R, Zhang X, Morelli J, Yan X, **Zhu X**, Li X., The Diagnostic Value of 3-Dimensional Sampling Perfection With Application Optimized Contrasts Using Different Flip Angle Evolutions (SPACE) MRI in Evaluating Lower Extremity Deep Venous Thrombus. [Invest Radiol. 2017 Jul 12](#)
 7. Xiang Z, Ai Z, Liang J, Li G, **Zhu X**, Yan X. Evaluation of Regional Variability and Measurement Reproducibility of Intravoxel Incoherent Motion Diffusion Weighted Imaging Using a Cardiac Stationary Phase Based ECG Trigger Method. [Biomed Res Int. 2018 Apr 17;2018:4604218](#)
 8. Xu X, Lin X, Huang J, Pan Z, **Zhu X**, Chen K, Zee CS, Yan F., The capability of inflow inversion recovery magnetic resonance compared to contrast-enhanced magnetic resonance in renal artery angiography. [Abdom Radiol \(NY\). 2017 May 3.](#)
 9. Zhang W, Jin X, Ding Y, Zhu X, Rong N, Li J, S. Composition of phosphorus in wetland soils determined by SMT and solution ³¹P-NMR analyses. [Environ Sci Pollut Res Int. 2016 May;23\(9\):9046-53](#)
 10. **Zhu X**, Chen S, Luo Q, Ye C, Liu M, Zhou X. Body temperature sensitive micelles for MRI enhancement. [Chem Commun \(Camb\).2015 Jun 4;51\(44\):9085-8.](#)

Poster & Presentation

1. Shujuan Li, Lixin Wang, Wei Yu, **Xiaolei Zhu**, Jingliang Cheng, Tianyi Qian. Shortening the Waiting Time for Primovist: Hepatobiliary Phase Enhancement Using Deep Learning Methods. Power ISMRM 2020
2. **Xiaolei Zhu**, Guijin Li, Zhiyong Li. Effectiveness of fatty fraction by high-speed T2-corrected multi-echo SVS (HISTO) and multiecho 3D DIXON (mDIXON) in rapid and robust acquisition for Duchenne Muscular Dystrophy. Proc. Intl. Soc. Mag. Reson. Med. 26 (2018): 3910
3. Zhiming Xiang, Zhu Ai, Guijin Li, **Xiaolei Zhu**, Xu Yan. A cardiac stationary phase based ECG trigger(CaspECG) of Intravoxel incoherent motion diffusion MR in left liver lobe. Proc. Intl. Soc. Mag. Reson. Med. 25 (2017): 5808

4. Xuehua Li, Siyun Huang, Mengchen Zhang, Siting Feng, **Xiaolei Zhu**, Xu Yan, Ziping Li. Magnetization transfer MRI for evaluating bowel fibrosis and inflammation in patients with stricturing Crohn's disease. Proc. Intl. Soc. Mag. Reson. Med. 25 (2017): 4490
5. Weiyuan Huang, Jijianjun Li, Feng Chen, Yingman Zhao, and **Xiaolei Zhu**. Prediction of Radiotherapy Response in Nasopharyngeal Carcinoma Patients Using Diffusion-Kurtosis Imaging. Proc. Intl. Soc. Mag. Reson. Med. 25 (2017): 1828
6. Weixiong Fan, Xiaofeng Chen, Fengyan Cheng, Yabao Cheng, Tai Xu, Guijin Li, **Xiaolei Zhu**, and Shuai Li. The Combine Analysis of improved Multi-parametric MR for identifying the different subtypes and immunohistochemical expression of Breast cancer. Proc. Intl. Soc. Mag. Reson. Med. 25 (2017): 5394
7. Gang Wu, Xu Yan, **Xiaolei Zhu**, Tianjing Zhang. The Diagnostic Value of Sampling Perfection Application with Contrast Optimized Using Variable Flip Angle Evolutions (SPACE MR) in Evaluating Lower Extremity Deep Venous Thrombus. Proc. Intl. Soc. Mag. Reson. Med. 25 (2017): 2817

PATENT

1. CHINESE PATENT: 201510063559.3. Zhou Xin; **Zhu Xiaolei**; Chen Shizhen; Sun Xianping; Liu Maili; [source](#)